

Supplementary Material:

Delphi-Psychometric Evaluation of 3D Generative Models

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1. Robustness Checks

E5: Leave-one-model-out cross-validation. We validated the tinyEval3D 30-item subset by holding out each model in turn, re-selecting items from the remaining five models using difficulty-stratified sampling, and measuring ranking recovery across all six held-out models. The difficulty-stratified strategy yields mean LOO-CV Spearman $\rho = 0.954$, indicating robust generalisation to unseen models. The IRT- θ -stratified strategy achieves the same LOO-CV $\rho = 0.954$, as prompt generatability scores are stable across this leave-one-out perturbation. Both substantially outperform what Fisher-information maximisation achieved under the old (unstable, $N = 6$ persons) IRT orientation, confirming that the flipped 2PL orientation is the key enabler.

E3, E4: Dichotomization threshold sensitivity. We re-ran E3 on geometric consistency with thresholds at 3, 4, 6, and 7 (out of 9), alongside the main threshold of 5. The proportion of zero-variance items is *not* monotonic in threshold: it is U-shaped (threshold=3: 48.2%, threshold=4: 31.2%, threshold=5: 7.8%, threshold=6: 9.4%, threshold=7: 81.4%), because a zero-variance item is one where all six models agree (all pass or all fail), which is common at extreme thresholds (almost everyone passes a very low bar, or almost everyone fails a very high one) and rare in the middle. Threshold=5 is close to the minimum of this curve, which is additional support for it as the main-text choice rather than an arbitrary midpoint. The difficulty-stratified subset strategy is robust to threshold choice as long as sufficient non-zero-variance items remain (threshold ≤ 6 recommended, based on this curve).

E2: Judge sensitivity. The judge facet accounts for only 1.7% of variance, and ICC values (0.81–0.88 per criterion) confirm high inter-judge reliability. Dropping any single judge shifts mean scores by less than 0.3 points on the 0–9 scale; model rankings are stable across all three leave-one-judge-out conditions (Spearman $\rho \geq 0.90$ relative to three-judge consensus).

E2: Modeling-choice robustness (fixed vs. random criterion; raw vs. standardized scale). Table 1 compares three specifications of the E2 variance decomposition on the automated-score stream: (i) the main-text model, with criterion as a fixed effect and model $\times$ criterion, prompt $\times$ criterion interactions included; (ii) an alternative treating criterion as a random facet (no criterion interactions, matching the simpler model used in an earlier draft); and (iii) the random-facet specification applied to per-criterion z -scored scores. All three agree that criterion (or its scale) dominates on the raw score, and that standardizing removes this by construction while reallocating variance mainly to prompt and model. Table 2 shows the same three specifications for the Delphi stream, where criterion’s share is already small on the raw scale (6–7%) because judges rate all criteria on a common 0–9 scale, so standardization changes the picture only modestly.

Table 1: Automated-score stream: % of total variance under three specifications. “—” marks a facet not present in that specification (fixed-criterion interactions are undefined under a random-criterion model, and vice versa). Column sums equal 100% in each specification.

Facet	Fixed criterion	Random criterion	Random criterion, z -scored
Criterion	82.0%	84.7%	0.0%
Residual	10.1%	11.4%	59.8%
Model $\times$ Criterion	2.1%	—	—
Model $\times$ Prompt	2.0%	1.2%	4.7%
Prompt $\times$ Criterion	1.5%	—	—
Model	1.4%	1.6%	14.8%
Prompt	0.9%	1.1%	20.7%
Total	100.0%	100.0%	100.0%

Table 2: Delphi stream: % of total variance under the same three specifications.

Facet	Fixed criterion	Random criterion	Random criterion, z -scored
Model	37.3%	37.4%	41.2%
Residual	16.9%	22.4%	22.1%
Model $\times$ Prompt	14.1%	13.6%	14.8%
Prompt	12.9%	13.9%	15.1%
Criterion	6.0%	6.6%	0.0%
Prompt $\times$ Criterion	5.4%	—	—
Round	4.5%	4.3%	4.7%
Judge	1.7%	1.8%	2.1%
Model $\times$ Criterion	1.3%	—	—
Total	100.0%	100.0%	100.0%

Notably, prompt $\times$ criterion is the fourth-largest facet in the Delphi stream (5.4%)—larger than criterion’s own main effect and than model $\times$ criterion (1.3%)—indicating that some prompts are differentially easy or hard depending on which criterion a judge is applying. This interaction was omitted from an earlier draft of the model with no stated justification; it does not change which facet dominates (model quality), but it is large enough that its exclusion should not be assumed harmless without checking, as we do here.

E2: The automated scorer as a fourth rater. To directly compare rater severity across measurement instruments, we treat the automated scorer as a fourth rater alongside the three Delphi judges, restricted to the 100 prompts $\times$ 5 models both streams cover and to judges’ Round 1 (independent, pre-anchor) scores to avoid circularity with the automated score already being shown to judges from Round 2 onward (main text, Delphi protocol). Because the automated scorer’s per-criterion ranges differ sharply from the judges’ uniform 0–9 scale (main text, E2), we z -score each criterion across all four raters before comparing severity or fitting any model—otherwise scale differences would masquerade as severity differences.

Table 3: Rater severity (mean z -score; more negative = harsher) and variance components with vs. without the automated scorer as a fourth rater. Criterion treated as a fixed effect, as in the main text’s Table 3.

	3 LLM judges only	+ automated (4 raters)
Gemini severity	−0.386	−0.322
Automated severity	—	−0.290
Claude severity	−0.100	−0.011
GPT severity	+0.487	+0.625
Model (% variance)	32.2%	23.6%
Residual (% variance)	25.7%	42.3%
Judge (% variance)	17.4%	17.5%

The automated scorer’s severity sits between Gemini and Claude—harsher than Claude, comparable to Gemini—so it is not a severity outlier relative to the LLM judges. Adding it leaves judge variance essentially unchanged (17.4%→17.5%), but residual variance rises sharply (25.7%→42.3%) while model variance falls correspondingly (32.2%→23.6%): the automated scorer’s disagreement with the LLM judges is not a fixed calibration offset (which would show up as judge variance) but idiosyncratic, item-level disagreement (residual)—a harder form of disagreement to correct for than a simple severity adjustment.

E1/E3: Pooled IRT across all four raters. We further pool all four raters’ person-level ratings—(model, prompt, rater) triples, not judge-averaged scores—into a single graded-response IRT model (criteria as items, as in E1’s MIRT), recoding each criterion onto a common 5-category scale via quantile bins computed across the pooled data so the differing native ranges do not distort item category structure. Within this single fitted model, splitting $\hat{\theta}$ by rater type reproduces the same split as the separate per-stream fits in the main text: automated-only $\hat{\theta}$ ranks HunyuanDiT-3D highest (−0.02) and TRELIS second (−0.16); Delphi-judge-only $\hat{\theta}$ ranks TRELIS highest (0.69) and HunyuanDiT-3D second (0.58). The pooled estimate (both rater types combined) narrowly favors TRELIS (0.475 vs. HunyuanDiT-3D’s 0.432), but each model receives 100 automated person-rows against roughly 300 Delphi-judge person-rows (three judges $\times$ 100 prompts), so the pooled estimate is not a neutral adjudication between the two methods—it simply reflects that Delphi contributes three times as many rated instances per model.

Delphi Protocol Flowchart

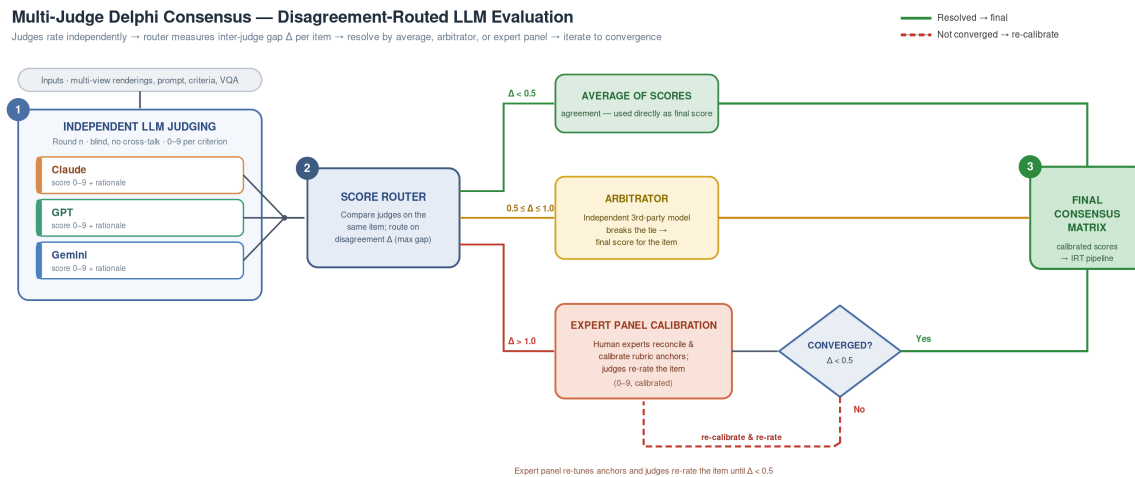


Figure 1: Delphi-structured LLM judging protocol. Three vision-capable LLM judges rate each (model, prompt) pair independently in Round 1, then receive anonymized group statistics and a dissent summary before re-rating in Rounds 2 and 3. Round 3 is skipped when inter-judge delta drops below 0.5 on ≥ 4 of 5 criteria after Round 2.